PABLO SABATER
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EDUCATION

Bachelor of Science in Electrical and Computer Engineering - Honors Program *Expected 05/2027*
Northeastern University - Boston, MA ***GPA: 3.75***

- ***Coursework:*** Circuits and Signals, Calculus 3, Linear Systems, Cornerstone of Engineering, Physics 2, Diff. Eq., Linear Algebra, Python, Electronics, Embedded/Digital Design, Networks, Algorithms (C++), Probability, Cybersecurity, Networks
- ***Honors:*** Honors Program / Honors Scholarship Recipient / Dean's List x5 / Honors Impact Badge
- ***Groups:*** Northeastern Electric Racing, Energy Efficient Circuits and Systems, IEEE-Eta Kappa Nu ECE Honor Society

TECHNICAL SKILLS

- ***Hardware:*** Oscilloscopes, Multimeters, Signals, PCB, High Voltage Power Electronics, Testing, Crimping
- ***Software:*** Azure, System Verilog, C++, Python, MATLAB, Git, Power BI, Excel, RStudio, SQL, LTSpice, ROS Noetic
- ***Systems:*** Linux, Windows, macOS, FPGA, Embedded Systems, Arduino, Microcontrollers
- ***Design:*** Altium Designer, KiCAD, Blender, AutoCAD, SolidWorks, 3D Printing, Laser Cutting, Soldering

ENGINEERING WORK EXPERIENCE

R&D Electrical and Computer Engineer Co-Op *07/2026-Present*
Miele Technologies Inc - Peabody, Massachusetts

- Investigating device high-voltage power **transients** using an industrial **programmable AC** power supply and **oscilloscope**
- Developed an **ROS**-based **robotics automation** system using **Python** to test and verify hardware over 100+ hours of run time; produced and conducted in depth **data analysis** using **Python Pandas** to validate up to 1% accuracy
- Constructed a **proof-of-concept** board for a voice monitoring system using **KiCAD**; manufactured in-house using **CNC milling** and PCB **reflow** and bench tested using multimeters and oscilloscopes
- Developing an **automation tool** using **Azure AI**, **agentic workflows, Playwright,** custom **MCP servers**, and **SQL datasets** used for internal analysis; presented to leadership with positive feedback

Microfluidics Hardware Co-Op *06/2025-12/2025*
Anthology Biotechnologies - Cambridge, Massachusetts

- Developed and provisionally **patented** an optical fungal screening and **robotic automation** platform from scratch
- Machined a custom platform designed using **SolidWorks** to house a custom **Arduino**-based **gantry** system that deposited and scanned in a custom **PDMS** enclosure
- Programmed a **Python** fluorescence scanning GUI using **Matplotlib** and **GRBL** to detect localized spore growth
- Led independent time-lapse imaging over 168 hours to systematically test the platform and suggest improvements

ENGINEERING CLUBS

Northeastern Electric Car Racing Club - Head of Electrical Powertrain *01/2024-Present*
Northeastern - Boston, Massachusetts

- Spearheading electrical design for the newest EV accumulator, owning key technical decisions like battery pack sizing by **simulating tradeoffs** in **MATLAB** based on previous competition **data**
- Leading the **flex-PCB** voltage and temperature pack sensing project; Redesigning custom **BMS** hardware to improve **cell balancing** circuits and IC reliability by ~20%
- Previous lead for **high-voltage PCB** designed in **Altium** that handles **relay** control, voltage monitoring; personally redesigned the battery **precharge/discharge** circuit from **analog** to **digital** control around ADBMS2950B IC
- Owned hardware bring-up of 3+ multi-revisioned **PCB** projects and validated reliability via **bench testing**
- Led and participated in 20+ **design reviews** ranging from documentation to PCB layout design for multiple project leads

OTHER WORK/PROJECTS

Gait Keeper - Insole Technology Project 0*5/2026-Present*

- Designed a **wearable insole** project with real-time gait analysis using a velostat/**flex PCB** resistor array for injury diagnosis
- Owned the implementation of power and USB-C charging electronics and **I2C** communication between an **IMU** and **ESP32**
- Building the **semi-rigid flex PCB** in **Altium** and handling validation through **bench testing** and **simulation**

Single-Cycle RISC-V CPU Project 0*1/2026-05/2026*

- Programmed a PYNQ **FPGA** board using **System Verilog** in **Vivado** to simulate **RISC-V** single-cycle architecture and calculate signed arithmetic; includes an instruction decoder, program counter, ROM, ALU, registers, and 5+ other
- Developed rigorous test programs for all combinations and results to prove functionality; documented across multiple reports

LANGUAGES: **English:** Native/Bilingual | **Spanish:** Native/Bilingual
INTERESTS: Basketball, travelling, running, cooking